

Effects of coffee consumption in chronic hepatitis C: a randomized controlled trial.

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BACKGROUND: Coffee is associated with a reduced risk of hepatocellular carcinoma in patients with chronic C hepatitis. This prospective trial was aimed at assessing the mechanisms underlying coffee-related protective effects. **METHODS:** Forty patients with chronic hepatitis C were randomized into two groups: the first consumed 4 cups of coffee/day for 30 days, while the second remained coffee "abstinent". At day 30, the groups were switched over for a second month. **RESULTS:** At baseline, aspartate aminotransferase and alanine aminotransferase were lower in patients drinking 3-5 (Group B) than 0-2 cups/day (Group A) (56 ± 6 vs $74 \pm 11/60 \pm 3$ vs 73 ± 7 U/L $p=0.05/p=0.04$, respectively). HCV-RNA levels were significantly higher in Group B [$(6.2 \pm 1.5) \times 10(5)$ vs $(3.9 \pm 1.0) \times 10(5)$ UI/mL, $p=0.05$]. During coffee intake, 8-hydroxydeoxyguanosine and collagen levels were significantly lower than during abstinence (15 ± 3 vs 44 ± 16 8-hydroxydeoxyguanosine/10(5)deoxyguanosine, $p=0.05$ and 56 ± 9 vs 86 ± 21 ng/mL, $p=0.04$). Telomere length was significantly higher in patients during coffee intake (0.68 ± 0.06 vs 0.48 ± 0.04 Arbitrary Units, $p=0.006$). Telomere length and 8-hydroxydeoxyguanosine were inversely correlated. **CONCLUSION:** In chronic hepatitis C coffee consumption induces a reduction in oxidative damage, correlated with increased telomere length and apoptosis, with lower collagen synthesis, factors that probably mediate the protection exerted by coffee with respect to disease progression. Copyright © 2012 Editrice Gastroenterologica Italiana S.r.l. Published by Elsevier Ltd. All rights reserved.